INCH-POUND

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6 July 2010
SUPERSEDING
MIL-DTL-82945A(OS)
21 January 1997

DETAIL SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT, CKU-10/A AND CKU-11/A ASSEMBLIES

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the CKU-10/A and CKU-11/A Rocket Catapults (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in Sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-46994	Pellets/Granules Boron/Potassium Nitrate
MIL-DTL-32124	Propellant, NSWC/IH-BC-15, NSWC/IH-BC-16, NSWC/IH-BC-17, NSWC/IH-BC-20, NSWC/IH-BC-21 and NSWC/IH-BC-22

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AMSC N/A FSC 1377

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1907	Inspection, Liquid Penetrant and Magnetic Particle, Soundness Requirements for Materials, Parts, and Weldments
DOD-STD-2101	Classification of Characteristics
MIL-STD-2102	Aircrew Escape Propulsion Systems; Vibration and Shock Tests for

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Surface Warfare Center, Indian Head Division (CAGE Code 14083)

DL 513-174-0100	Catapult, Aircraft Ejection Seat, CKU-10/A Assembly
DL 513-174-0200	Catapult, Aircraft Ejection Seat, CKU-11/A Assembly

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E11G7, Document Control, 4072 North Jackson Road Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes
ANSI/ASQ Z1.9	Sampling Procedures and Tables for Inspection by Variables for Percent Nonconforming

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(Copies of this document are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1417	Standard Practice for Liquid Penetrant Testing
ASTM E 1742	Standard Practice for Radiographic Examination.

(Copies of these documents are available online at http://www.astm.org or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE AS 17983	Compass, Magnetic, Pilot's Standby

(Copies of these documents are available on line at http://www.sae.org or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.1.1 Pressure transducers. All first article sample catapult assemblies shall be adapted for pressure transducers as specified in the contract or order (see 6.2 and 6.4).

3.1.2 Propellant aging. Propellant samples shall be prepared and tested in accordance with the requirements of 4.3.3.

3.2 Materials. All materials, parts, and assemblies shall be in accordance with the applicable DL and this specification.

3.2.1 Primary components. Only primary components from a single primary component lot shall be used in a production lot of loaded rocket catapults.

3.2.2 Propellant. The rocket motor propellant shall be in accordance with MIL-DTL-32124, Type V and the catapult propellant shall be in accordance with MIL-DTL-32124, Type VI.

3.2.3 Material Definitions.

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3.2.3.1 Propellant batch. A propellant batch is a mix of ingredients, each ingredient composed of one homogeneous lot that is mixed at one time in one vessel, resulting in a homogeneous mixture.

3.2.3.2 Rocket motor grain assembly batch. A rocket motor grain assembly batch is a group of rocket motor grain assemblies cast from one propellant batch.

3.2.3.3 Rocket motor grain assembly lot. A rocket motor grain assembly lot is a group of rocket motor grain assemblies composed of one or more rocket motor grain assembly batches all having the same ingredient lots.

3.2.3.4 Catapult grain assembly batch. A catapult grain assembly batch is a group of catapult grain assemblies cast from one propellant batch.

3.2.3.5 Catapult grain assembly lot. A catapult grain assembly lot is a group of catapult grain assemblies composed of one or more catapult grain assembly batches all having the same ingredients lots.

3.2.3.6 Rocket catapult lot. A rocket catapult lot is a group of units composed of one lot of rocket motor grain assemblies, one lot of catapult grain assemblies, one lot of external igniter assemblies, and one lot of rocket motor igniters that are assembled at one time.

3.2.3.7 External igniter assembly lot. An external igniter assembly lot is a group of external igniter assemblies that are composed of one lot of percussion primers, one lot of $BKNO_3$ pellets, and one lot of $BKNO_3$ granules that are assembled at one time.

3.2.3.8 Rocket motor igniter lot. A rocket motor igniter lot is a group of external igniter assemblies that are composed of one lot of $BKNO_3$ pellets and one lot of $BKNO_3$ granules that are assembled at one time.

3.2.3.9 Primary components. Primary components are all components containing explosive ingredients. These include the percussion primers, the external igniter assembly, the catapult grain assembly, the rocket motor grain assembly, and the rocket motor igniter assembly.

3.2.3.10 Major metal parts. The major metal parts are the motor tube (Drawing 513-174-0101), the nozzle (Drawing 513-174-0102 or Drawing 513-174-0201), the motor tube head (Drawing 513-174-0110), the double tang lock (Drawing 513-174-0219), the breech (Drawing 513-174-0133), the launcher tube (Drawing 513-174-0146), and the trunnion (Drawing 513-174-0147).

3.3 Age of explosive components. The following paragraphs give the requirements for controlling the age of explosive components.

3.3.1 Propellant. The propellant is defined as the rocket motor grain (rocket grain assembly per Drawing 513-174-0217) and the catapult grain (catapult grain assembly per Drawing 513-174-0218). For purposes of this section, loaded is defined as when the rocket catapult lot is released for lot acceptance testing.

3.3.1.1 Rocket motor grain. The rocket motor grain shall be loaded within 180 days of the oldest rocket motor propellant mix.

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3.3.1.2 Catapult grain. The catapult grain shall be loaded within 180 days of the oldest catapult propellant mix.

3.3.2 Other explosive components. The other explosive components are the percussion primer (Drawing 8799925), the $BKNO_3$ pellets (MIL-P-46994, Type II-A), and the $BKNO_3$ granules (MIL-P-46944, Type VII, 40/140 granulation).

3.3.2.1 Percussion primers. The percussion primers shall be loaded within 24 months of manufacture and certification or within 12 months of recertification. Loaded is defined as when the external igniter assembly (Drawing 513-174-0123) is complete.

3.3.2.2 $BKNO_3$ pellets and granules. The $BKNO_3$ pellets and granules shall be loaded into the rocket motor igniter (Drawing 513-174-0107) within 12 months of the certification date. Loaded is defined as when the rocket motor igniter is complete. The $BKNO_3$ pellets and granules shall be loaded into the cartridge case assembly (Drawing 513-174-0119) within 12 months of the certification date. Loaded is defined as when the cartridge case assembly is crimped.

3.3.2.3 External igniter assembly and rocket motor igniter assembly. The external igniter assembly (Drawing 513-174-0123) and the rocket motor igniter (Drawing 513-174-0107) shall be loaded into the rocket catapult within 6 months of their completion. Loaded is defined as when the rocket catapult lot is released for lot acceptance testing.

3.4 Performance and product characteristics. The rocket catapults shall meet the following performance and product characteristics when test fired at -65°F, 70°F, and 165°F using an ejected weight of 347 ± 1 pounds.

3.4.1 Ballistics. The rocket catapults shall meet the ballistic performance parameters specified in table I.

3.4.1.1 Ballistic Definitions. See figure 1 for a graphical representation of some of the definitions given below.

3.4.1.1.1 Rocket motor resultant impulse. The rocket motor resultant impulse is obtained by integrating the resultant thrust of the rocket motor over the action time.

3.4.1.1.2 Rocket motor action time. Action time is determined from the normal thrust-time curve, and is defined as the time interval from the 10 percent of maximum thrust on the initial rise of curve to the corresponding 10 percent of thrust on the declining portion of the curve.

3.4.1.1.3 Rocket motor resultant thrust. The rocket motor resultant thrust is obtained by vectorially adding the two maximum components of rocket thrust that occur at one time.

3.4.1.1.4 Rocket motor ignition delay. Rocket motor ignition time is the elapsed time from separation to 10 percent of maximum thrust, as measured on the rising portion of the normal thrust-time curve.

3.4.1.1.5 Catapult ignition delay time. Catapult ignition delay time is the elapsed time from the first indication of shearing of the firing pins until the first indication of catapult thrust.

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3.4.1.1.6 Catapult separation velocity. Catapult separation velocity is defined as the average velocity of the simulated ejected weight propelled horizontally at catapult separation measured over the interval from separation to 12 inches after separation.

3.4.1.1.7 Catapult dynamic response index (DRI). The DRI is representative of the maximum dynamic compression of the vertebral column of the human body. In physical terms, the DRI is calculated by mathematically describing the human body in terms of an analogous, lumped parameter mechanical model consisting of a mass, spring, and damper. The DRI is determined from the equations (1) and (2).

$$\frac{d^2\delta}{dt} + 2\zeta\omega_n \bullet \frac{d\delta}{dt} + \omega_n{}^2\delta = \frac{d^2z}{dt^2} \quad (1)$$

$$DRI = \frac{\omega_n{}^2\delta_{\max}}{g} \quad (2)$$

Where:

δ = compression of spring
ζ = 0.224 (damping ration of the model)
ω_n = 52.9 radians/sec (undamped natural frequency of the model)
$\frac{d^2z}{dt^2}$ = catapult acceleration

Substituting given numerical values, these equations become:

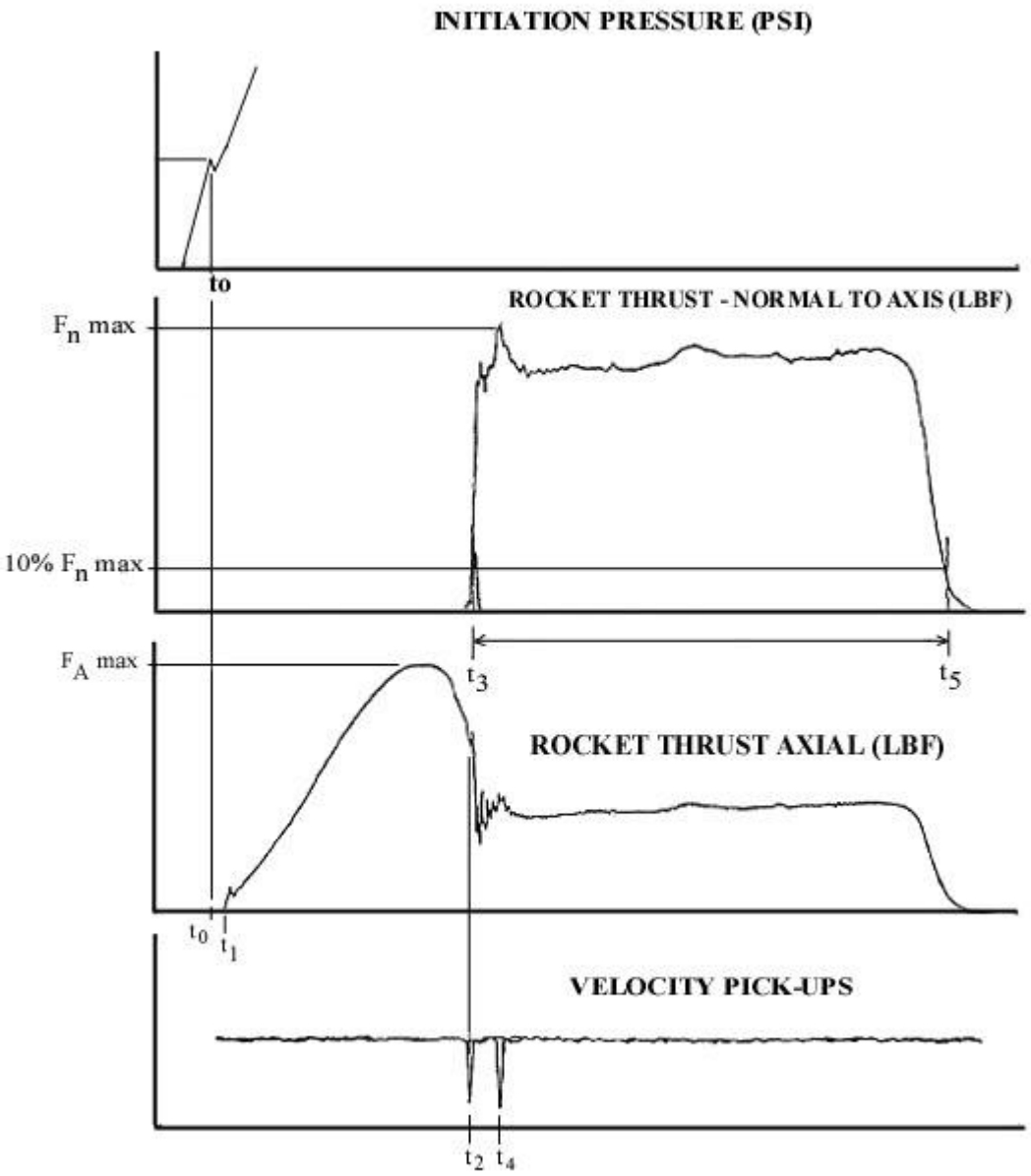
$$\frac{d^2\delta}{dt^2} + 23.7\frac{d\delta}{dt} + 2798\delta = \frac{d^2z}{dt^2}$$

$$DRI = 86.9\delta_{\max}$$

3.4.1.1.8 Catapult launch time. Catapult launch time is the time between the first indication of thrust on the thrust-time curve until separation.

3.4.1.1.9 Catapult axial thrust. The maximum thrust from the thrust-time curve that occurs during the catapult launch time.

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$t_1 - t_0$ = **Catapult Ignition Delay Time**

$\frac{1\,foot}{t_4 - t_2}$ = **Catapult Separation Velocity**

F_A max = **Catapult Axial Thrust-maximum**

F_n max = **Rocket Motor Normal (vertical) Thrust-maximum**

$t_3 - t_2$ = **Rocket Motor Ignition Delay**

$t_5 - t_3$ = **Rocket Motor Action Time**

FIGURE 1. Performance Curves

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TABLE I. Ballistic performance requirements for a 347-pound ejected weight.

Parameter	-65°F		70°F		165°F		AQL
	Min	Max	Min	Max	Min	Max	
Catapult							
Initiation pressure, (psi)	350	550	350	550	350	550	—
Dynamic response index, (DRI)[1/]	—	18.0	—	18.0	—	22.0	0.65
Separation velocity, (ft/s)[2/]	33	—	33	—	33	—	0.65
Ignition delay, (ms)	—	10	—	10	—	10	0.65
Axial thrust (lb_f)	—	7000	—	7000	—	7000	1.0
Launch time (ms)	—	260	—	260	—	260	1.0
Rocket Motor							
Resultant impulse, (lb_f/sec)	960	—	960	—	960	—	1.0
Ignition delay, (ms)	—	30	—	30	—	30	1.0
Action time, (ms)[3/]	—	535	—	535	—	535	1.0
Resultant thrust, (lb_f)	—	5000	—	5000	—	5000	0.65
Thrust angle (reference)* CKU-10/A CKU-11/A	 47.5° 54.5°		 47.5° 54.5°		 47.5° 54.5°		 — —

1/ DRI is defined in section 3.4.1.1.7.
2/ Nominal separation velocities are: 37.0 ft/s at -65°F and 46.0 ft/s at 165°F.
3/ Nominal rocket motor action time is: 470 ms at -65°F and 340 ms at 165°F.
* Record for information only.

3.4.1.2 Other requirements. In addition to meeting the ballistic requirements, none of the following shall occur during test firing of rocket catapults:

a. No unit component, including but not limited to catapult components, O-rings, springs, or pieces of propellant shall be ejected.

b. No component subject to loading shall fail.

c. No excessive nozzle erosion, hot spots, or burnthrough shall occur.

d. No leakage of ballistic gas past a sealing surface shall occur.

e. The thrust-time curve shall exhibit a shape that compares well with the established (or representative) curve for the unit.

3.4.2 Initiation. Maximum allowed initiation pressure input is 10,000 psi. The rate of pressurization input shall be 30,000 to 50,000 psi/sec.

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3.5 Hydrostatic pressure. The components listed in table II shall withstand the internal hydrostatic pressure specified for 15 seconds minimum when tested in accordance with 4.4.6.2.

TABLE II. Hydrostatic pressure requirements.

Component	Drawing	Hydrostatic pressure (psi)
Motor Tube	513-174-0101	5,000
Launcher Tube	513-174-0146	1,500
Breech Assembly	513-174-0143	1,500

3.6 Leak test. Each head assembly shall pass the leak test specified in 4.4.6.5.

3.7 Radiographic. Each loaded rocket catapult shall be examined radiographically to ascertain the presence, condition, and correct placement of internal components.

3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.9 Residual magnetism. The rocket catapult assembly, when passed in front of an SAE AS 17983 compass along the entire length of the catapult (see figure 2), shall not cause the compass indicator to deflect more than 5 degrees in either direction. This test shall be performed in an area free from local magnetic effects.

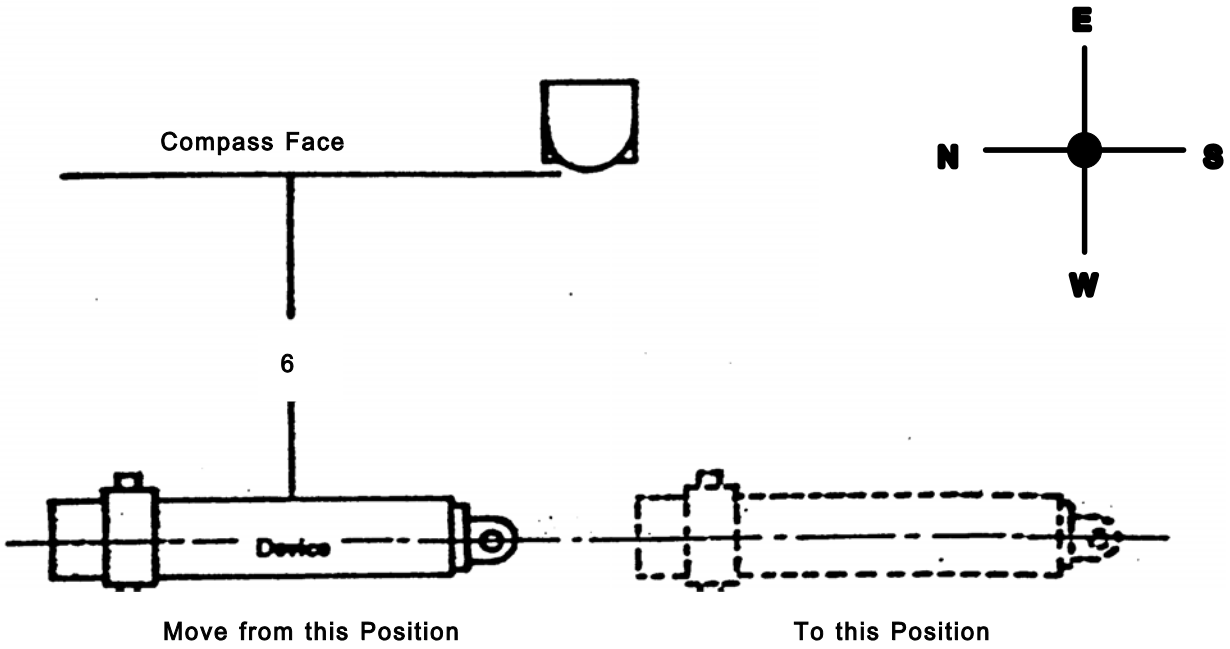
3.10 Environmental. The rocket catapult assemblies shall withstand the environmental conditioning of 3.10.1 through 3.10.4. Rocket catapult assemblies subjected to this conditioning shall show no evidence of deterioration of propellant or disarrangement of components when examined in accordance with 4.4.6.1 and shall meet the ballistic performance requirements of 3.4 when test fired in accordance with 4.3.2.

3.10.1 High temperature. The rocket catapult shall withstand the high temperature test of 4.3.1.1.

3.10.2 Vibration. The rocket catapult shall withstand the vibration test of 4.3.1.2.

3.10.3 Temperature-shock. The rocket catapult shall withstand the temperature-shock test of 4.3.1.3.

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NOTE: Device and compass shall lie in a common horizontal plane

FIGURE 2. Layout for determination of residual magnetism in device.

3.10.4 Thirty-day storage test. The rocket catapult shall withstand the 30-day storage test of 4.3.1.4.

3.11 Identification markings. Each CKU-10/A rocket catapult shall be marked in accordance with Drawing 513-174-0100 with an identification label, a warning label, a nozzle label, an external igniter label, and an installation label. Each CKU-11/A rocket catapult shall be marked in accordance with Drawing 513-174-0200 with an identification label, a warning label, a nozzle label, an external igniter label, and an installation label.

3.11.1 Production lot designation. The production lot number shall follow MIL-STD-1168. The construction of the lot number shall be as follows:

XXX	10	A	001	-001
a	b	c	d	e

Where:

a = Manufacturer's identification code.
b = The year of manufacture of the oldest batch of propellant (catapult or rocket motor) in the lot.

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c = The month of manufacture of the oldest batch of propellant (catapult or rocket motor) in the lot expressed as an alpha code in accordance with 4.1.3 of MIL-STD-1168.
d = Lot interfix number (controlled by Indian Head Division, Naval Surface Warfare Center, Code E28, Indian Head, MD 20640-5000 or procuring activity).
e = Lot sequence number - lot number (manufacturer is responsible for the assignment of lot numbers.)

3.11.2 Major metal parts serialization. Each major metal part shall have a nonrecurring serial number permanently etched on the part. The serial number shall provide traceability to the drawing revision the part was manufactured to, the metal part manufacturer and contract number, and the raw material and heat lots.

3.12 Workmanship. The workmanship shall be that required by the applicable drawings, referenced specifications, and the best industrial practices governing the quality production of interchangeable parts for maintaining the dimensions, finishes, tolerances, and quality specified herein. All components and assemblies shall be free from burrs, contamination, corrosion, sharp edges, or foreign material which could result in malfunction of the unit or components, or be a safety hazard in handling.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient 65 to 95°F
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.3 First article inspection. Unless otherwise specified in the contract or order (see 6.2), a first article sample shall consist of three blocks of rocket motor propellant, three blocks of catapult propellant, two rocket motor grain assemblies, two catapult grain assemblies, and eighteen loaded rocket catapult assemblies. All first article samples shall be manufactured using the methods and procedures proposed for production. The first article samples shall be subjected to the tests of 4.4, with the exception of the test firing of 4.4.5, followed by environmental testing, radiographic examination, and static firing in the sequence specified in table III. Acceptance of the first article sample shall be based on no defects in the sample. Any production of the rocket catapult assemblies by the contractor prior to approval of the first article samples shall be at the contractor's risk. First article samples shall not be included as part of the quantity specified for delivery in the contract.

4.3.1 Environmental tests.

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4.3.1.1 High temperature. The rocket catapult assembly shall be subjected to the high-temperature storage test in accordance with Method 501 of MIL-STD-810. The test items shall be maintained at 165 ± 5°F for 50 hours minimum. Operation of the rocket catapult shall be only as specified in table III.

4.3.1.2 Vibration. The rocket catapult assembly shall be subjected to random vibration for jet aircraft in accordance with MIL-STD-2102 with the exception that the duration shall be for 6 hours, with 2 hours on each axis. Before vibration, units shall be temperature conditioned in accordance with table III for a minimum of 8 hours. The input vibratory acceleration must have a W_o factor of 0.1 g^2/Hz and the reduction factor must be 0 g^2/Hz-dB. If the rocket catapult must be removed from the environmental conditioning chamber or if the temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall be stopped and the unit reconditioned. A reconditioning time of 2 minutes for every minute outside the specified temperature envelope shall be required. A continuous temperature history of the unit will be kept for all conditioning and testing time. Unless otherwise noted, the rocket catapult shall be mounted on a test fixture in a manner that will simulate actual seat/aircraft installation and excited through the attachment fittings. In addition to compliance with the random vibration test, the rocket catapult shall comply with the gun fire vibration test of Method 519 of MIL-STD-810 for those applications whereby the guns are installed on the aircraft (see 6.2). However, if the maximum test spectrum level of the gun fire configuration is determined to be equal to or less than the MIL-STD-2102 test level, the gun fire vibration test shall not be required.

4.3.1.3 Temperature shock. The rocket catapult assembly shall be subjected to the temperature shock test in accordance with Method 503 of MIL-STD-810. The test temperature extremes shall be -65 ± 5°F and 165 ± 5°F and the duration of exposure at each temperature shall be 8 hours. Operation of the rocket catapult shall be only as specified in table III.

4.3.1.4 Thirty-day storage. Three rocket catapults shall be placed in storage at -65°F and three at 165°F for 30 days each. Proper temperature shall be maintained throughout storage.

4.3.2 Static firing test. Within 30 days after completion of environmental exposure the rocket catapults shall be static fired at the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5000, Attn: Code E28, for aging testing. The sample shall be temperature conditioned per table III for a minimum of 8 hours and test fired within 10 minutes after removal from the conditioning chamber to verify conformance to the requirements of 3.4. Data specified in table I shall be analyzed in accordance with ANSI/ASQ Z1.9, Section B, Part I, using the procedures of B-5 through B-7. Nonconformance with table I or any requirement of 3.4.1 shall be cause for rejection of the first article sample.

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TABLE III. First article evaluation test program.[1/]

Test Para.	Test	Unit Number																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
4.4.6.1	Radiographic inspection	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a
4.3.1.1	High temperature	b	b	b	b	b	b												
4.3.1.2	Vibration -65°F							b	b	b									
4.3.1.2	Vibration 165°F										b	b	b						
4.3.1.4	30-day storage -65°F													b	b	b			
4.3.1.4	30-day storage 165°F																b	b	b
4.4.6.1	Radiographic inspection	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c
4.3.1.3	Temperature shock	d	d	d	d	d	d	d	d	d	d	d	d						
4.4.6.1	Radiographic inspection	e	e	e	e	e	e	e	e	e	e	e	e						
4.3.2	Ballistic test -65°F	f	f					f			f			d			d		
4.3.2	Ballistic test 70°F			f	f				f			f			d			d	
4.3.2	Ballistic test 165°F					f	f			f			f			d			d

1/ Letters denote sequence of testing.

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4.3.3 Propellant aging. Three blocks of cured propellant from each batch of rocket motor propellant and three blocks from each batch of catapult propellant used in the first article sample shall be submitted to the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5000, Attn: Code E28, for aging testing. High-temperature aging shall be conducted for 84 days with an atmosphere of 22 percent relative humidity and at 165 ± 5°F. At zero time and at the end of the 28-, 56-, and 84-day aging, physical properties shall conform to MIL-DTL-32124, Type V (rocket motor propellant) and MIL-DTL-32124, Type VI (catapult propellant). The rate for stress/strain determination shall be 2.0 inches/minute. The sample shall be cast into rectangular cardboard containers approximately 1.5 inches high by 8 inches wide by 11 inches long. All samples submitted shall be bare propellant, individually wrapped in moistureproof packaging and properly identified as to propellant lot and batch number. Failure of the propellant to meet the requirements of the applicable specification shall be cause for rejection of the propellant batch.

4.3.4 Propellant morphology. Two rocket motor grain assemblies and two catapult grain assemblies from each batch of propellant used in the first article sample shall be submitted for morphological examination of the propellant. All samples shall be individually wrapped in moistureproof packaging and properly identified as to propellant lot, batch number, and grain serial number. These tests shall be used as baseline information.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests of 4.4.4 through 4.4.6, conducted on samples selected as specified in 4.4.2. The contractor is responsible for assuring that all rocket catapults offered for acceptance meet all requirements of the applicable drawings and specifications. Items of inspection shall be serialized and inspection data traceable to a specific serial number.

4.4.1 Test reports. When specified in the contract or order (see 6.2), the contractor shall furnish test reports to the procuring activity. Test reports shall include the following information as described in 4.4.1.1 through 4.4.1.7:

a. Propellant description (see 3.2.2)
b. Static firing performance test results (see 4.4.5.1)
c. Summary sheet
d. Ammunition data cards
e. Propellant/assembly production records
f. Certification of age of explosive components and propellant (see 3.3)
g. Metal parts manufacturing records

4.4.1.1 Propellant description. The following propellant information shall be submitted with each production lot. The materials and components of each propellant batch shall list the manufacturer, lot number, and date of manufacture. The burning rates for the rocket motor propellant at 1000, 2000, 3000, and 4000 psi performed at -65°F and 165°F for each pressure (three samples at each temperature and pressure per batch) shall be included. The burning rates for the catapult propellant at 500, 1000, 1500, and 2000 psi performed at 77°F for each pressure (three samples at each temperature and pressure per batch) shall be included. In addition, include the results of tensile strength and elongation tests at -65°F, 77°F, and 165°F with a pull rate of 2 inches/minute (five samples at each temperature per batch), heat of

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explosion (three samples per batch), and verification that ingredients were properly weighed and added to each batch (mix sheet).

4.4.1.2 Static firing performance tests. All computations and data required for ballistic evaluation generated during quality conformance inspection static firing per 4.4.5 (Government) and 4.4.5.1 (contractor) shall be reported, including firing traces. The following information shall also be listed:

a. Grain designation, manufacturer, lot number, and date of manufacture
b. Type of test equipment
c. Deviations from standard testing procedures
d. Additional comments (malfunctions, unusual occurrences)
e. Disassembly findings report (see 4.4.5.2)

4.4.1.3 Summary sheet. A summary of the results of the inspections required by 3.5 through 3.9 shall be submitted to the designated Government activity for each inspection lot. The summary shall include the sample size, number and nature of defects found, and the disposition (accepted or rejected).

4.4.1.4 Data cards. Data cards (DD Form 1650) shall be completed for each loaded rocket catapult lot. Copies shall be provided in accordance with contract requirements and one copy shall be forwarded to the Commander, Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5000 (Code E28).

4.4.1.5 Propellant and assembly production records. The contractor shall prepare propellant production records and retain them for the service life of the unit. These records shall include batch number, batch designation, rocket motor grain assembly lot number, rocket motor grain assembly serial number, catapult grain assembly lot number, catapult grain assembly serial number, external igniter assembly lot numbers and serial numbers, and rocket motor igniter lot number. The records shall be cross referenced to the assembled rocket catapult lot number and to specific rocket catapult serial numbers.

4.4.1.6 Age certification. A certification of age of explosive components and propellants shall be furnished by the contractor to show compliance with section 3.

4.4.1.7 Major metal parts manufacturing records. For each major metal part, the contractor shall prepare and retain for the service life of the unit, production records that link the serial number of the metal part with the drawing revision, manufacturer and contract number, and raw material and heat lots for that part. The serial number of these metal parts shall be cross referenced to the assembled rocket catapult lot number and to the specific rocket catapult serial number.

4.4.2 Sampling for quality conformance inspection.

4.4.2.1 Lot. For the purposes of identification, inspection, and shipment, a lot of loaded assemblies shall consist of a maximum of 420 assemblies, including test samples. The homogeneity of inspection lots shall be maintained as follows:

a. Components were produced by homogeneous process.

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b. Units are loaded with same lots of rocket motor grain assemblies, catapult grain assemblies, external igniter assemblies, and rocket motor igniters (see 3.2.3.3, 3.2.3.5, 3.2.3.7, and 3.2.3.8).

4.4.2.2 Sampling for examination. Designation of defects (critical, major, or minor) shall be in accordance with DOD-STD-2101.

a. Critical and major defects: Each rocket catapult assembly (Drawing 513-174-0100 or Drawing 513-174-0200) shall be examined for all critical and major defects (see 4.4.4.1).

b. Minor defects: Sample rocket motor catapult assemblies (Drawing 513-174-0100 or Drawing 513-174-0200) submitted for examination shall be selected in accordance with general inspection level II of ANSI/ASQC Z1.4, AQL 2.5 (see 4.4.4.2).

c. Components: Unless a specific contractor-generated inspection plan for all subcontracted components has been approved by the procuring activity (see 4.4.4.3), the following sampling for components shall apply. All rocket catapult components shall be examined for all critical defects. Sampling for other defects shall be in accordance with level II of ANSI/ASQC Z1.4, AQL 1.0 for major defects and AQL 2.5 for minor defects.

4.4.2.3 Sampling for firing test. A sample of loaded rocket catapult assemblies shall be submitted to the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5000, for static firing in accordance with 4.4.5. The sample size shall be in accordance with table IV. Samples shall be chosen by Indian Head Division, Naval Surface Warfare Center, Indian Head, MD personnel or by a designated Government representative upon completion of the production lot assembly. Loaded rocket catapult samples shall not be preselected during production. Rocket catapults selected for samples shall not be included as part of the quantity specified for delivery in the contract or order. The ballistic sample size shall be divided as equally as possible among rocket motor batches and catapult batches. From each rocket motor batch or catapult batch not represented in the sample rocket catapult assemblies, one sample rocket motor or catapult grain shall be selected for static firing by the contractor (see 4.4.5.1), so that at least one rocket motor or catapult grain from each batch has been static fired.

4.4.2.4 Sampling for nondestructive test. Each rocket catapult assembly shall be subjected to the tests specified in 4.4.6.

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TABLE IV. Sample size for firing test.

Lot size	Total firing sample	Sample for temperature extreme
Up to 110	10	5
111 - 180	14	7
181 - 300	20	10
301 - 340	22	11
341 - 380	24	12
381 - 420	26	13

4.4.3 Resubmission of rejected lot. A sample of rocket catapult assemblies shall be selected in accordance with 4.4.2.3 and submitted when, with prior approval of the procuring activity, the contractor requests that the rejected lot be test fired under different conditions or the contractor believes a rejected lot may be reworked to meet the requirements of this specification.

4.4.4 Examination.

4.4.4.1 Critical and major defects. Each rocket catapult assembly shall be examined for critical and major defects in accordance with the applicable drawings, 3.5, 3.7, and 3.9, and the sampling plan of 4.4.2.2.a. Any rocket catapult assembly found to contain a critical or major defect shall be rejected.

4.4.4.2 Minor defects. Sample rocket catapult assemblies selected in accordance with 4.4.2.2 b. shall be examined for minor defects in accordance with the applicable drawings, 3.8, 3.11, and 3.12. If any rocket catapult assembly is found to contain a minor defect, the rocket catapult lot shall be rejected. The lot may be resubmitted for examination of minor defects when all of the defective units are removed from the lot or all of the defects are corrected.

4.4.4.3 Component examination. Unless an alternative method is proposed, the rocket catapult components shall be examined in accordance with the applicable drawings and the sampling plan of 4.4.2.2 c. Any component found to contain a critical defect shall be rejected. If any component is found to contain a major or minor defect, the component lot shall be rejected. The component lot may be resubmitted for examination of major or minor defects when all of the defective units are removed from the lot or all of the defects are corrected. Alternative methods may be proposed and used with the approval of the procuring activity. The method shall be depicted in the contractor's quality program plan (see 6.2) and shall ensure that all fabricating and processing (of any type) are accomplished under controlled conditions which are monitored and accepted by the procuring activity. Systematic control is required for all facets of work to assure all dimensions and requirements are met. This control can be accomplished through direct or indirect inspection, process monitoring, or statistical process control, or a combination of all four. Inspection for rubber O-rings shall be in accordance with MIL-STD-413.

4.4.5 Production lot acceptance firing. Rocket catapults selected in accordance with 4.4.2.3 shall be static fired at the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5000, within 30 days after submission for acceptance of the rocket catapult lot. The sample catapults shall be

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divided into two equal groups with one group conditioned at -65 ± 5°F and the other group conditioned at 165 ± 5°F. Each group shall be conditioned for a minimum of 8 hours. The catapults shall be fired within 10 minutes after removal from the conditioning chamber. Minimum equipment accuracies shall be as shown in table V. Data specified in table I shall be analyzed in accordance with ANSI/ASQ Z1.9, Section B, Part I, using the procedures of B-5 through B-7. Nonconformance with table I or any requirement of 3.4 shall be cause for rejection of the lot.

TABLE V. Test equipment accuracies.

Measured Characteristic	Equipment Accuracy
Maximum thrust	± 2.0 percent
Time	± 0.001 second
Maximum acceleration	± 5 percent
Pressure	± 2.0 percent
Temperature	± 2.0°F
Thrust frequency response, minimum	60 Hz

4.4.5.1 Contractor batch acceptance test firing. Units (rocket motor grain assemblies or catapult grain assemblies) selected by the contractor to meet the requirements of 4.4.2.3 shall be static fired by the contractor as batch acceptance tests. The sample units shall be divided into two equal groups with one group conditioned at -65 ± 5°F and one group at 165 ± 5°F. Each group shall be conditioned for a minimum of 8 hours. The units shall be fired within 10 minutes after removal from the conditioning chamber. Minimum equipment accuracies shall be as shown in table V. Failure of any unit to meet the requirements of 3.4 shall be cause for rejection of the specific batch of rocket motor grain assemblies or catapult grain assemblies.

4.4.5.2 Post static-firing disassembly. A post firing disassembly and inspection shall be conducted on all first article, lot acceptance, and batch acceptance test units. Detailed observations on conditions of individual components, including marginal conditions, shall be noted. All findings shall be evaluated against the requirements of 3.4.1.2. When specified in the contract or order (see 6.2), a disassembly findings report shall be submitted to the contracting activity.

4.4.6 Nondestructive tests. Each rocket catapult assembly shall be subjected to the following tests. When specified in the contract (see 6.2), the contractor shall furnish test reports detailing the results of all nondestructive tests for each lot of rocket catapults.

4.4.6.1 Radiographic inspection. Rocket catapults shall be radiographically inspected in accordance with ASTM E 1742, radiographic quality level 2-1T. Any deficiency shall be cause for rejection of the unit. Radiographs shall be capable of detecting defects and rocket catapult components 0.03 inch in length or width. When specified in the contract (see 6.2), all radiographs taken of the rocket catapult assembly or components thereof shall be forwarded to the activity designated by the contracting activity. Otherwise,

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all radiographs shall be inspected on-site by a designated Government representative. Acceptance of the loaded rocket catapult lot shall be subject to approval of the radiographs by the designated Government activity.

4.4.6.2 Hydrostatic pressure. Each component listed shall be subjected to the respective internal hydrostatic pressure specified in table II for a minimum of 15 seconds without application of an external load. Any leakage, permanent deformation outside drawing tolerances, or mechanical failure shall result in rejection of that component. Each motor tube and launcher tube shall then be subjected to inspection of 4.4.6.3.

4.4.6.3 Penetrant inspection. Each motor tube, each launcher tube (prior to anodizing), and each unlock piston (prior to dry film lubrication) shall be 100 percent penetrant inspected in accordance with ASTM E 1417, Type I, Method D, Sensitivity Level 3. Discontinuities exceeding one-half the maximum size permitted by MIL-STD-1907, Grade A, Table I, are unacceptable.

4.4.6.4 Residual magnetism. Each rocket catapult assembly shall be passed 6 inches in front of an SAE AS 17983 compass along the entire length of the catapult in accordance with figure 2. If the compass indicator deflects more than five degrees in either direction the rocket catapult assembly shall be rejected.

4.4.6.5 Leak test. Each head assembly shall be subjected to a leak test. The MS33649-04 port shall be pressurized with dry nitrogen to 3000 to 3500 psi and there shall be no greater than a 5 percent loss in pressure within 15 seconds when measured at the 0.437 to 0.439 diameter outlet (ballistic tube interface).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements will be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The CKU-10/A and CKU-11/A Rocket Catapults are intended to be used to propel an aircraft ejection seat and its occupant safely out and away from a jet aircraft. Since these Rocket Catapults were designed for use in military aircraft, there are no commercial applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

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b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

c. Whether a first article sample is required (see 3.1, 4.3 and 6.3).

d. Instructions for adaption of first article samples for pressure transducers (see 3.1.1 and 6.4).

e. Assigned activity for first article inspection and static firing tests (see 4.3, 4.3.2, 4.4.2.3, 4.4.5, and 6.5).

f. Whether the gun fire vibration test is required (see 4.3.1.2).

g. Whether test reports are required (see 4.4.1, 4.4.5.2, and 4.4.6).

h. When radiographs are to be furnished to the Government (see 4.4.6.1).

i. Packaging requirements (see 5.1).

j. Disposition of hardware, scrap propellant, and metal parts (see 6.7).

k. Inspection conditions, if other than as specified (see 4.2).

l. Whether any of the following are required and, if so the appropriate DID: test reports, ammunition data cards, certification data, or product assurance quality program plan.

m. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" OP 5 are applicable.

n. Guidance for the serialization of the external igniter assembly (Drawing 513-174-0123) and the major metal parts (see 3.11.2).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Pressure transducers. Instructions for adapting the first article catapult assemblies for pressure transducers should be compatible with instruments having a response not less than 600 Hz and that are capable of detecting ignition peak pressures and unstable burning rates.

6.5 Test facility. The Indian Head Division, Naval Surface Warfare Center, Indian Head, MD is the design agent for the product covered by this specification. As such, they are equipped to perform the first article inspection and static firing tests specified herein. Except when special conditions warrant otherwise, the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD should be designated as the test facility for the first article inspection and the production acceptance test firings.

6.6 Explosive hazard classification. The explosive hazard classification for this rocket catapult has been designated as:

Explosive Hazard Classification – Class 1, Division 3
Storage Compatibility Group – Group C
Department of Transportation (DOT) Hazard Class – Class B
DOT Marking – Rocket Motors

6.6.1 Hazard notice. The rocket catapult described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.7 Disposition of hardware, scrap propellant, and metal parts. The contract should specify that:

a. all items purchased become the property of the U.S. Government

b. no hardware, propellant, or metal parts should be disposed of without authorization of the procuring activity

c. all items be made available to the procuring activity upon request

d. the disposition of any tested hardware, propellant, or metal parts without authorization by the procuring activity will result in the test being declared invalid, and a retest will be required.

6.8 Retention of radiographs. All radiographs of accepted lots of rocket catapults are to be retained on file by the contractor for 8 years or 1 year past the service life of the unit, whichever is longer.

6.9 Service life. The rocket catapult has a design life of 7 years and an installed service life of 7 years from the date of propellant manufacture.

6.10 Subject term (keyword) listing.

Rocket catapult
Rocket motor

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6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project 1377-2010-001)

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